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McMaster University

Mathematics and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 19
Undetermined coefficients
Tuesday 27 October 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Method of Undetermined Coefficients

[Trench §5.4], [Trench §5.5], [Trench §9.3].

Solving nonhomogeneous linear ODEs with constant coefficients.

Solutions of nonhomogeneous linear ODEs

Theorem (General Solution of Nonhomogeneous linear ODE [Trench 9.1.6])

Suppose y_p is any particular solution of the nonhomogeneous n^{th} order linear ODE, $L[y] = g(x)$, on an interval $\mathcal{I}$, and suppose $\{y_1, y_2, \ldots, y_n\}$ is a fundamental set of solutions for the associated homogeneous equation, $L[y] = 0$, on $\mathcal{I}$. Then the ***general solution*** *of $L[y] = g(x)$ on the interval $\mathcal{I}$ is*

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x),$$

where c_i, $i = 1, 2, \ldots, n$, are arbitrary constants.

Note: $y_c(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x)$ is called the ***complementary function*** for the equation $L[y] = g(x)$. The general solution is $y = y_c + y_p$, where y_p is any solution of $L[y] = g(x)$.

Solutions of nonhomogeneous linear ODEs

Important notes for the case of ***constant coefficients*** in $L[y]$:

- If $L[y]$ has constant coefficients, then these coefficients are (trivially) continuous functions on $(-\infty, \infty)$.
- $\therefore$ $L[y] = 0$ has a general solution (y_c) that is valid on all of $(-\infty, \infty)$.
- $\therefore$ If $g(x)$ is continuous on an interval $\mathcal{I}$, then $L[y] = g(x)$ has a general solution ($y = y_c + y_p$) that is valid on $\mathcal{I}$.
- When can we find a formula for the general solution of $L[y] = g(x)$?
  - We must first find y_c, which we can do if we can solve the characteristic equation for $L[y] = 0$.
  - We must then find a particular solution y_p.
- Can we always find a formula for a particular solution y_p?
  - *Answer:* Yes. We'll now discuss important cases where we can find such a formula in terms of familiar functions.
    Note: In general, y_p can be expressed only in terms of an integral (Appendix A).

Solving $L[y] = g(x)$ via undetermined coefficients

ODEs that arise in practice often involve a function $g(x)$ that can be constructed from sums and products of

- constants (*e.g.*, 1, π, 7);
- polynomials (*e.g.*, x, x^3, $x^5 + 4x^2 + 12$);
- exponentials (*e.g.*, e^x, e^{-3x}, $e^{\alpha x}$);
- sines and cosines (*e.g.*, $\sin \beta x$, $\cos \gamma x$).

For example:

$$g(x) = 27, \qquad g(x) = x^2 + 2x + 1,$$

$$g(x) = (27x^{27} + 9x^9 + 3x^3)e^{4x} \cos 9x + 8x^6 e^{2x} \sin 5x$$

Functions of this general type have the property that their derivatives are also functions of this general type.

So, maybe y_p can be derived as a linear combination of $g(x)$ and its derivatives?

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = x^2 - 1$)

First solve the associated homogeneous equation, $L[y] = 0$:

$$y'' - 2y' + y = 0 \implies r^2 - 2r + 1 = 0 \qquad \textit{characteristic equation}$$

$$\implies (r-1)^2 = 0 \implies r = 1 \qquad \textit{repeated root}$$

$$\implies \{e^x, x\,e^x\} \qquad \textit{fundamental solution set}$$

$$\implies y_c = (c_1 + c_2 x)e^x \qquad \textit{complementary function}$$

Next, to find y_p, note that

$$g(x) = x^2 - 1, \qquad g'(x) = 2x, \qquad g''(x) = 2.$$

$\therefore$ Linear combinations of derivatives of $g(x)$ will be quadratic, so assume

$$y_p = Ax^2 + Bx + C, \qquad A, B, C \in \mathbb{R}.$$

$$\implies y_p' = 2Ax + B,$$

$$\implies y_p'' = 2A$$

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = x^2 - 1$ (continued))

$$\begin{aligned} \implies L[y_p] &= y_p'' - 2y_p' + y_p \\ &= (2A) - 2(2Ax + B) + (Ax^2 + Bx + C) \\ &= Ax^2 + (B - 4A)x + 2A - 2B + C \\ \text{set equal to } g(x): \quad &= x^2 \qquad\qquad - 1 \end{aligned}$$

$$\implies \left\{ \begin{aligned} A &= 1 \\ B - 4A &= 0 \\ 2A - 2B + C &= -1 \end{aligned} \right\} \implies \left\{ \begin{aligned} A &= 1 \\ B &= 4A = 4 \cdot 1 = 4 \\ C &= -1 - (2A - 2B) \\ &= -1 - (2 \cdot 1 - 2 \cdot 4) = 5 \end{aligned} \right\}$$

$\implies y_p = x^2 + 4x + 5$ *particular solution!*

$\implies y = y_c + y_p = (c_1 + c_2 x)e^x + x^2 + 4x + 5$ *general solution!*

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = 2\sin(x)\cos(x)$)

First solve the associated homogeneous equation, $L[y] = 0$:
Already done! (same homogeneous case as previous example)

$$y_c = (c_1 + c_2 x)e^x \qquad \textit{complementary function}$$

Can we apply the method of undetermined coefficients?

Yes, $g(x)$ is a product of the constant 2, $\sin(x)$, and $\cos(x)$.
It will be simpler to note first that $g(x) = \sin(2x)$ (by trig identity).
Next, to find y_p, note that

$$g(x) = \sin(2x), \qquad g'(x) = 2\cos(2x), \qquad g''(x) = -4\sin(2x)$$

$\therefore$ Assume y_p is a linear combination of $\sin(2x)$ and $\cos(2x)$:

$$y_p = A\sin(2x) + B\cos(2x), \qquad A, B \in \mathbb{R}.$$
$$\implies y_p' = 2A\cos(2x) - 2B\sin(2x)$$
$$\implies y_p'' = -4A\sin(2x) - 4B\cos(2x)$$

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = 2\sin(x)\cos(x)$ (continued))

$$
\begin{aligned}
\Longrightarrow \quad L[y_p] &= y_p'' - 2y_p' + y_p \\
&= (-4A\sin(2x) - 4B\cos(2x)) \\
&\qquad - 2(2A\cos(2x) - 2B\sin(2x)) \\
&\qquad\quad + (A\sin(2x) + B\cos(2x)) \\
&= (-4A + 4B + A)\sin(2x) + (-4B - 4A + B)\cos(2x) \\
&= (-3A + 4B)\sin(2x) + (-3B - 4A)\cos(2x) \\
\Longrightarrow \quad & -3A + 4B = 1 \quad \text{and} \quad -3B - 4A = 0 \\
\Longrightarrow \quad & -3A + 4B = 1 \quad \text{and} \quad B = -\frac{4}{3}A \\
\Longrightarrow \quad & -3A + 4\left(-\frac{4}{3}A\right) = 1 \quad \text{and} \quad B = -\frac{4}{3}A \\
\Longrightarrow \quad & -\frac{25}{3}A = 1 \quad \text{and} \quad B = -\frac{4}{3}A
\end{aligned}
$$

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = 2\sin(x)\cos(x)$ (continued))

$$\Longrightarrow \quad A = -\frac{3}{25} \quad \text{and} \quad B = \frac{4}{25}$$

$\Longrightarrow$ *particular solution:*

$$y_p = -\frac{3}{25}\sin(2x) + \frac{4}{25}\cos(2x)$$

$\Longrightarrow$ *general solution:*

$$\begin{aligned} y &= y_c + y_p \\ &= (c_1 + c_2 x)e^x - \frac{3}{25}\sin(2x) + \frac{4}{25}\cos(2x) \end{aligned}$$

Solving $L[y] = g(x)$ via undetermined coefficients

Example ($y'' - 2y' + y = 2\sin(x)\cos(x) - x^2 + 1$)

Note: RHS $g(x)$ is more complicated than previous two examples.

But note also that

$$2\sin(x)\cos(x) - x^2 + 1 = \sin(2x) - (x^2 - 1)$$

so $g(x)$ is the difference of the RHSs in the previous two examples.

Is this a useful observation?

Consider this question in the following poll.

Poll

- Go to
  https://www.childsmath.ca/childsa/forms/main_login.php
- Click on Math 2C03
- Click on Take Class Poll
- Fill in poll **Superposition of particular solutions**
- Submit.

Solving $L[y] = g(x)$ via undetermined coefficients

Example $(y'' - 2y' + y = 2\sin(x)\cos(x) - x^2 + 1)$

From the previous examples, we know that

$$L[x^2 + 4x + 5] = x^2 - 1$$

$$L\Big[-\tfrac{3}{25}\sin(2x) + \tfrac{4}{25}\cos(2x)\Big] = \sin(2x)$$

Therefore, since L is a linear operator, for any y_1 and y_2,

$$L[y_1 - y_2] = L[y_1] - L[y_2],$$

$$\therefore L\Big[-\tfrac{3}{25}\sin(2x) + \tfrac{4}{25}\cos(2x) - (x^2 + 4x + 5)\Big] = \sin(2x) - (x^2 - 1)$$

$$\begin{aligned} \Longrightarrow \quad y &= y_c + y_p \\ &= (c_1 + c_2 x)e^x - \tfrac{3}{25}\sin(2x) + \tfrac{4}{25}\cos(2x) - (x^2 + 4x + 5) \end{aligned}$$

$L[y] = g(x)$ subject to initial or boundary conditions

Very often, rather than the most general solution, we want the solution to an ***initial value problem (IVP)***:

$$\text{Solve} \quad L[y] = g(x) \quad \text{subject to} \quad y(x_0) = y_0,\ y'(x_0) = y_1$$

Note: We have assumed here that L is *2nd order*. For n^{th} order ODEs, we need n initial conditions.

In other situations, specific conditions are specified at different points. For example, if L is 2nd order, and $x_0 < x_1$:

$$\text{Solve} \quad L[y] = g(x) \quad \text{subject to} \quad y(x_0) = y_0,\ y(x_1) = y_1,$$

Here, both conditions specify values of the unknown function y, not its derivatives. But the two values are specified at different points, x_0 and x_1. This is a ***boundary value problem (BVP)***.

$L[y] = g(x)$ Initial Value Problem (IVP)

Example ($y'' - 2y' + y = x^2 - 1, \quad y(0) = 1,\ y'(0) = 1$)

We found the general solution:

$$y(x) = (c_1 + c_2 x)e^x + x^2 + 4x + 5$$

$$\begin{aligned} \Longrightarrow \quad y'(x) &= c_2 e^x + (c_1 + c_2 x)e^x + 2x + 4 \\ &= [c_1 + c_2(x+1)]e^x + 2x + 4 \end{aligned}$$

Inserting the initial conditions:

$$y(0) = c_1 + 5 = 1 \quad \Longrightarrow \quad c_1 = -4$$

$$y'(0) = c_1 + c_2 + 4 = 1 \quad \Longrightarrow \quad -4 + c_2 + 4 = 1 \quad \Longrightarrow \quad c_2 = 1$$

$\therefore$ The unique solution to the IVP is

$$y(x) = (x - 4)e^x + x^2 + 4x + 5$$

$L[y] = g(x)$ Boundary Value Problem (BVP)

Example ($y'' - 2y' + y = x^2 - 1, \quad y(0) = 1,\ y(1) = 1$)

We found the general solution:

$$y(x) = (c_1 + c_2 x)e^x + x^2 + 4x + 5$$

Inserting the boundary conditions:

$$\begin{aligned} y(0) &= c_1 + 5 = 1 \quad \Longrightarrow \quad c_1 = -4 \\ y(1) &= (c_1 + c_2 1)e^1 + 1 + 4 + 5 = (c_1 + c_2)e + 10 = 1 \\ &\Longrightarrow \quad c_1 + c_2 = -9/e \quad \Longrightarrow \quad c_2 = 4 - 9/e \end{aligned}$$

$\therefore$ The BVP is solved by

$$\begin{aligned} y(x) &= [-4 + (4 - 9/e)x]e^x + x^2 + 4x + 5 \\ &= (-4 + 4x)e^x - \tfrac{9}{e}xe^x + x^2 + 4x + 5 \\ &= 4(x-1)e^x - 9xe^{x-1} + x^2 + 4x + 5 \end{aligned}$$

$L[y] = g(x)$ Boundary Value Problem (BVP)

In the previous example, the boundary conditions specified the value of the function $y(x)$ at two points. But we could have specified values of $y'(x)$ at those points instead. There are many possible forms of boundary conditions. See Appendix B for general linear boundary conditions, their matrix form, and nonlinear conditions.

Note: BVPs need not have unique solutions.

- Some BVPs do have unique solutions.
- Some BVPs have infinitely many solutions.

  Note: If there are two distinct solutions then there are infinitely many: if y_1 and y_2 solve the same linear BVP, then $ty_1 + (1-t)y_2$ is also a solution for every $t \in \mathbb{R}$.
- Some BVPs have no solutions at all!

[Trench §13.1] discusses boundary value problems.

Solving $L[y] = g(x)$ via undetermined coefficients

Examples of RHSs $g(x)$ and forms of y_p to try:

	$g(x)$	Form of y_p
1.	1 (any constant)	A
2.	$5x + 7$	$Ax + B$
3.	$3x^2 - 2$	$Ax^2 + Bx + C$
4.	$x^3 - x + 1$	$Ax^3 + Bx^2 + Cx + E$
5.	$\sin 4x$	$A\cos 4x + B\sin 4x$
6.	$\cos 4x$	$A\cos 4x + B\sin 4x$
7.	e^{5x}	Ae^{5x}
8.	$(9x - 2)e^{5x}$	$(Ax + B)e^{5x}$
9.	x^2e^{5x}	$(Ax^2 + Bx + C)e^{5x}$
10.	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
11.	$5x^2\sin 4x$	$(Ax^2 + Bx + C)\cos 4x + (Ex^2 + Fx + G)\sin 4x$
12.	$xe^{3x}\cos 4x$	$(Ax + B)e^{3x}\cos 4x + (Cx + E)e^{3x}\sin 4x$

Solving $L[y] = g(x)$ via undetermined coefficients

Note: The method of undetermined coefficients fails if the proposed particular solution y_p contains terms that occur in the complementary function y_c.

- If this happens, use the idea that allows us to solve $L[y] = 0$ when the characteristic equation has repeated roots.
- *e.g.*, If $g(x)$ has a term proportional to e^x and y_c also has such a term, then try a term of the form $A\,x\,e^x$ in the proposed y_p.
  - If y_c also has a term proportional to $x\,e^x$, then try a term of the form $A\,x^2\,e^x$ in the proposed y_p.
    - If y_c also has a term proportional to $x^2\,e^x$, then try a term of the form $A\,x^3\,e^x$ in the proposed y_p.

      ...
- *General rule:* If the trial for one group of terms in $g(x)$ overlaps with y_c, then multiply the whole trial for that group by x^n, where n is the smallest positive integer that eliminates the duplication.

Practice problems

- Polynomial and exponential forcing
  [Trench §5.4]:
  - Exercises 1, 4, 7, 11: particular solutions.
  - Exercises 15, 18: general solutions; 20, 22: solve the IVPs.
  - Exercises 24, 26: superposition.
  - Exercises 31–36: substitute the proposed forms directly.
- Sines and cosines, with polynomial or exponential factors
  [Trench §5.5]: Exercises 1, 3, 7, 10, 12, 17.
- Higher-order equations
  [Trench §9.3]: Exercises 1, 8, 12, 33, 60, 64.

Appendix A: deriving a general integral formula

Consider

$$L[y] \;=\; a_2(x)y'' + a_1(x)y' + a_0(x)y \;=\; g(x),$$

with continuous coefficients and g on $\mathcal{I}$, and $a_2(x) \neq 0$. Let $\{y_1, y_2\}$ be a fundamental solution set for $L[y] = 0$ on $\mathcal{I}$.

- Replace the constants in $c_1y_1 + c_2y_2$ by functions:

$$y_p \;=\; u_1y_1 + u_2y_2.$$

- The product rule gives

$$y_p' \;=\; u_1'y_1 + u_2'y_2 + u_1y_1' + u_2y_2'.$$

  We have two functions to choose. Impose $u_1'y_1 + u_2'y_2 = 0$, so that

$$y_p' \;=\; u_1y_1' + u_2y_2', \qquad y_p'' \;=\; u_1'y_1' + u_2'y_2' + u_1y_1'' + u_2y_2''.$$

- Substitution into $L[y_p] = g$ now gives

$$u_1\underbrace{L[y_1]}_{0} + u_2\underbrace{L[y_2]}_{0} + a_2(u_1'y_1' + u_2'y_2') \;=\; g.$$

This is ***variation of parameters***. Reference: W. F. Trench, *Elementary Differential Equations*, free ed. (2013), §5.7, equations (5.7.3)–(5.7.9). His P_0, P_1, P_2, F are our a_2, a_1, a_0, g.

Continue to the integral formula

Appendix A: the general integral formula

The preceding calculation requires

$$\begin{aligned} y_1u_1' + y_2u_2' &= 0, \\ y_1'u_1' + y_2'u_2' &= g/a_2, \end{aligned} \qquad \text{i.e.,} \quad \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ g/a_2 \end{pmatrix}.$$

Its determinant is the Wronskian $W = y_1y_2' - y_1'y_2$, which is nonzero on $\mathcal{I}$.

Multiply the first equation by y_2', the second by y_2, and subtract:

$$(y_1y_2' - y_1'y_2)u_1' = -y_2g/a_2, \qquad \text{i.e.,} \quad Wu_1' = -y_2g/a_2.$$

Similarly, multiply the second equation by y_1, the first by y_1', and subtract:

$$(y_1y_2' - y_1'y_2)u_2' = y_1g/a_2, \qquad \text{i.e.,} \quad Wu_2' = y_1g/a_2.$$

Integrate from any $x_0 \in \mathcal{I}$, then substitute into $y_p = u_1y_1 + u_2y_2$:

$$\boxed{y_p(x) = -y_1(x)\int_{x_0}^{x} \frac{y_2(t)g(t)}{a_2(t)W(t)}\,dt + y_2(x)\int_{x_0}^{x} \frac{y_1(t)g(t)}{a_2(t)W(t)}\,dt.}$$

The integrands are continuous, so these integrals exist for every $x \in \mathcal{I}$. This gives a particular solution even when the integrals cannot be evaluated using familiar functions.

Reference: [Trench §5.7, equations (5.7.9)–(5.7.12)].

Higher-order version Return to the main lecture

Appendix A: the integral formula for order n

Suppose $L[y] = a_n(x)y^{(n)} + \cdots + a_0(x)y = g(x)$ is normal on $\mathcal{I}$, and $\{y_1, \ldots, y_n\}$ is a fundamental solution set. Seek $y_p = \sum_{j=1}^n u_j y_j$.

- Choose the u_j' by solving

  $$\sum_{j=1}^n u_j' y_j^{(k)} = 0 \quad (0 \le k \le n-2), \qquad \sum_{j=1}^n u_j' y_j^{(n-1)} = \frac{g}{a_n}.$$

  The coefficient matrix is the Wronskian matrix; its determinant is nonzero. Thus this linear system uniquely determines continuous functions $u_1', \ldots, u_n'$.

- Repeated use of the product rule and the first $n-1$ conditions gives

  $$y_p^{(k)} = \sum_{j=1}^n u_j y_j^{(k)} \quad (0 \le k \le n-1).$$

  Differentiating once more and substituting into L gives

  $$L[y_p] = \sum_{j=1}^n u_j \underbrace{L[y_j]}_{0} + a_n \sum_{j=1}^n u_j' y_j^{(n-1)} = g.$$

- Write $q_j(x)$ for the computed $u_j'(x)$. We obtain

  $$\boxed{y_p(x) = \sum_{j=1}^n y_j(x) \int_{x_0}^x q_j(t)\,dt}, \qquad x_0 \in \mathcal{I}.$$

Reference: [Trench §9.4, equations (9.4.2)–(9.4.6)]. His $P_0, \ldots, P_n, F$ are our $a_n, \ldots, a_0, g$.

Return to the main lecture

Appendix B: general linear two-point boundary conditions

For a second-order ODE on $[x_0, x_1]$, each linear boundary condition may involve all four endpoint values:

$$\alpha_j y(x_0) + \beta_j y'(x_0) + \gamma_j y(x_1) + \delta_j y'(x_1) \;=\; b_j, \qquad j = 1, 2.$$

Define the column vector $\mathbf{Y}(x) = (y(x), y'(x))^{\mathsf{T}}$. The two conditions can be written

$$A\mathbf{Y}(x_0) + B\mathbf{Y}(x_1) \;=\; \mathbf{b},$$

where

$$A = \begin{pmatrix} \alpha_1 & \beta_1 \\ \alpha_2 & \beta_2 \end{pmatrix}, \quad B = \begin{pmatrix} \gamma_1 & \delta_1 \\ \gamma_2 & \delta_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

- For two independent conditions, require rank $\begin{pmatrix} A & B \end{pmatrix} = 2$: the two rows of this 2×4 matrix must be linearly independent.
- Neither A nor B has to be invertible. This rank condition alone does not guarantee a unique BVP solution.

Reference: U. M. Ascher, R. M. M. Mattheij and R. D. Russell, *Numerical Solution of Boundary Value Problems for Ordinary Differential Equations*, SIAM (1995), §1.1.3. doi:10.1137/1.9781611971231.

Examples of boundary conditions

Appendix B: separated and coupled boundary conditions

In the matrix form $A\mathbf{Y}(x_0) + B\mathbf{Y}(x_1) = \mathbf{b}$:

- ***Separated conditions*** impose one condition at each endpoint:

$$a_j y(x_j) + b_j y'(x_j) = c_j, \quad j = 0, 1, \qquad (a_j, b_j) \neq (0, 0).$$

Thus

$$A = \begin{pmatrix} a_0 & b_0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ a_1 & b_1 \end{pmatrix}.$$

Prescribing y gives ***Dirichlet*** conditions; prescribing y' gives ***Neumann*** conditions; a combination gives ***Robin*** conditions. The two endpoints may use different types.

- ***Coupled conditions*** connect values at different endpoints. For example, ***periodic*** conditions are

$$y(x_0) = y(x_1), \qquad y'(x_0) = y'(x_1),$$
$$A = I_2, \quad B = -I_2, \quad \mathbf{b} = \mathbf{0}.$$

Here I_2 is the 2×2 identity matrix. ***Antiperiodic*** conditions instead have minus signs on the RHSs, so $A = B = I_2$.

Reference: Ascher, Mattheij and Russell (1995), §1.1.3.

When is the solution unique?

Appendix B: existence and uniqueness for linear BVPs

Suppose $L[y] = g$ is normal on an interval containing $[x_0, x_1]$. Using a fundamental set $\{y_1, y_2\}$, write the general solution as $y = y_p + c_1 y_1 + c_2 y_2$. In column-vector form,

$$\mathbf{Y}(x) = \mathbf{Y}_p(x) + H(x)\mathbf{c}, \qquad H(x) = \begin{pmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}.$$

Inserting this into the boundary conditions gives

$$\underbrace{\big(AH(x_0) + BH(x_1)\big)}_{K}\mathbf{c} = \underbrace{\mathbf{b} - A\mathbf{Y}_p(x_0) - B\mathbf{Y}_p(x_1)}_{\mathbf{d}}.$$

Thus the BVP reduces to the linear system $K\mathbf{c} = \mathbf{d}$.

- If $\det K \neq 0$, there is exactly one solution.
- If $\det K = 0$, the system either has no solution or infinitely many. It has solutions precisely when $\operatorname{rank} K = \operatorname{rank} \begin{pmatrix} K & \mathbf{d} \end{pmatrix}$.

This explains why independence of the boundary conditions alone is insufficient: the matrix K also depends on the ODE.

Reference: Ascher, Mattheij and Russell (1995), §3.1.1.

Higher-order boundary conditions

Appendix B: boundary conditions for order n

For an nth-order ODE, collect the endpoint data in

$$\mathbf{Y}(x) = \begin{pmatrix} y(x) & y'(x) & \cdots & y^{(n-1)}(x) \end{pmatrix}^{\mathsf{T}}.$$

The general set of n linear two-point conditions is

$$A\mathbf{Y}(x_0) + B\mathbf{Y}(x_1) = \mathbf{b}, \qquad A, B \in \mathbb{R}^{n\times n}, \quad \mathbf{b} \in \mathbb{R}^n.$$

- Independence of these n conditions means rank $\begin{pmatrix} A & B \end{pmatrix} = n$. Neither matrix individually has to be invertible.
- For a normal linear ODE, form the $n \times n$ matrix $H(x)$ whose jth column is $(y_j(x), y_j'(x), \ldots, y_j^{(n-1)}(x))^{\mathsf{T}}$, using a fundamental set.
- The same argument as for second order reduces the BVP to

  $$K\mathbf{c} = \mathbf{d}, \qquad K = AH(x_0) + BH(x_1).$$

  An invertible K gives one solution. Otherwise, there are either no solutions or infinitely many, according to consistency.

Reference: Ascher, Mattheij and Russell (1995), §1.1.3 and §3.1.1.

Nonlinear boundary conditions

Appendix B: nonlinear boundary conditions

Even for a linear ODE, two-point boundary conditions may be nonlinear:

$$B_j\big(y(x_0), y'(x_0), y(x_1), y'(x_1)\big) \;=\; 0, \qquad j = 1, 2.$$

The functions B_j need not be linear combinations of their arguments. The number of solutions can now be *finite and greater than one*.

- For example, consider the BVP

  $$y'' \;=\; 0 \quad (0 \le x \le 1), \qquad y'(0) \;=\; 0, \qquad y(1)^2 \;=\; 1.$$

  The ODE gives $y(x) = c_1 + c_2 x$. The first condition gives $c_2 = 0$; the second then gives $c_1^2 = 1$. There are exactly two solutions:

  $$\boxed{y(x) \equiv 1 \quad \text{or} \quad y(x) \equiv -1}$$

- Their average $y \equiv 0$ still solves the ODE, but fails the nonlinear condition $y(1)^2 = 1$. The argument using $ty_1 + (1-t)y_2$ therefore does not apply.
- For any integer $m \ge 1$, keeping $y'(0) = 0$ and imposing $\prod_{j=1}^{m}(y(1) - j) = 0$ gives exactly m solutions. Using $\sin(\pi y(1)) = 0$ gives $y \equiv k$ for every integer k: a countably infinite set.

Reference for the general boundary-condition formulation: Ascher, Mattheij and Russell (1995), §1.1.3.

Linear boundary conditions Return to the main lecture